

Zamad Shakeel

hello@zamdevai.com | linkedin.com/in/zamad-gopang | github.com/zamadshakil | zamdevai.com | Lahore, Pakistan

Professional Summary

AI/SaaS Engineer and Technical Lead with 3+ years of experience shipping 12+ production-grade software products. Specialized in full-stack development, AI agent architectures (LangChain, LangGraph, CrewAI), and scalable SaaS platforms built on Python, Next.js, and PostgreSQL. Proven track record of building hybrid search engines, multi-model LLM orchestration systems, and enterprise ERP platforms. Lead Organizer of Google Developer Groups on Campus (GDGoC) at University of Central Punjab.

Skills

Languages: Python, JavaScript, TypeScript, SQL, HTML, CSS, C++

AI/ML: LangChain, LangGraph, CrewAI, AutoGen, RAG Pipelines, Vector Databases (pgvector, Pinecone), Sentence Transformers, BM25

Frameworks: Next.js, React, FastAPI, Numpy, Django, Flask, Node.js

Databases: PostgreSQL, PostGIS, Supabase, Redis, MongoDB

Cloud & DevOps: Docker, Kubernetes, Amazon Web Services (AWS), Google Cloud Platform (GCP), DigitalOcean, Vercel, Cloudflare, CI/CD, GitHub Actions

Tools & Platforms: Celery, WebSockets, Stripe API, Playwright, Git, REST APIs, GraphQL, Nginx

Architecture: Microservices, Event-Driven Design, RBAC, Row-Level Security (RLS), Async Job Queues, Real-Time Systems, Hybrid Search

Work Experience

Founder & Technical Lead

Apr 2024 – Present

ZamDev AI — Boutique AI Engineering Studio

Lahore, Pakistan

- Shipped 12+ production-grade SaaS products for founders and startups, delivering MVPs in 3-week sprint cycles using Next.js, FastAPI, PostgreSQL, and Stripe
- Architected multi-model LLM orchestration system routing dynamically based on task complexity, achieving sub-200ms latency with Docker/Kubernetes
- Built tools.zamdevai.com — a 35+ free developer tools platform (Python WebAssembly compiler, SEO auditor, flow diagramming) with 100% client-side execution and PWA offline support
- Designed and deployed end-to-end CI/CD pipelines, automated Playwright testing suites, and database performance optimization (EXPLAIN ANALYZE indexing) for client production systems

Senior Web Developer

Jul 2024 – Apr 2025

XITAD — Enterprise IT Asset Disposition

Remote

- Engineered a custom Enterprise Resource Planning (ERP) system for the IT Asset Disposition (ITAD) industry, replacing manual spreadsheet workflows with a unified digital audit trail
- Implemented immutable chain-of-custody tracking ledger ensuring 100% compliance with GDPR and HIPAA data privacy regulations across hardware asset lifecycle
- Built real-time inventory dashboard using WebSockets displaying warehouse levels, asset grades, and resale values with live synchronization across distributed teams
- Developed automated PDF certification engine generating “Certificates of Destruction” upon workflow completion, reducing administrative overhead by 90%
- Designed granular Role-Based Access Control (RBAC) with permission systems for Logistics Drivers, Warehouse Managers, and Auditors

Web Developer (Contract)

Jul 2024 – Feb 2025

Super Mini Print Shop — E-Commerce Automation

Remote

- Created automated file validation pipeline checking uploaded artwork for resolution (DPI) and dimensions before order acceptance, reducing print errors to zero
- Engineered dynamic pricing algorithm calculating costs in real-time based on square footage, material type, and bulk quantity breakpoints
- Built custom SinaLite API integration routing line items to third-party wholesalers for fulfillment while keeping internal production items in-house

Founder & Sole Developer

Oct 2023 – Aug 2024

WriteAI.me — AI Writing SaaS Platform

Remote

- Designed and built an AI content generation platform with 80+ writing templates, an AI Document Editor (Monaco), and autonomous Agent Mode for long-form content
- Architected backend routing system (Python/FastAPI) dynamically switching between multiple LLMs based on cost optimization, reducing API spend by 50%
- Deployed stateless containerized infrastructure (Docker/Kubernetes) handling concurrent requests with sub-200ms latency
- Integrated Stripe Subscription API with tiered usage quotas, automated dunning management, and real-time usage tracking

Projects

- Hierarchy — AI Document Review for Inspection & Audit** | showcase.system.zamdevai.com 2025
- Built AI-powered document validation system that reviews inspection reports against quality rules in under 60 seconds, saving managers 6–10 hours per week of manual report reading
 - Implemented 17-module platform with Smart AI (RAG-based chat via pgvector), role-based access control, validation rules engine, and real-time activity logging using Next.js and Supabase
 - Achieved 82% AI pass rate with average score of 84.2 across 23 weekly submissions in production environment
- ZamDev AI Tools Platform** | tools.zamdevai.com | *Next.js, WebAssembly, PWA* 2026
- Built 35+ free developer tools including Python compiler (Pyodide/WebAssembly + Monaco editor), SEO & Performance Auditor (Google PageSpeed API), and ZamDev Flow diagramming tool
 - Engineered 100% client-side architecture — zero server uploads, zero sign-ups, instant browser execution with PWA offline support via service workers
 - Published 3 technical blog articles (Base64 security, SVG vs PNG performance, JSON debugging) driving organic developer traffic
- HYDRA — Hybrid Dense-and-Sparse Search Engine** | hydra.zamdevai.com | *FastAPI, PostgreSQL, pgvector* 2025
- Architected a hybrid search engine combining BM25 keyword search with dense vector semantic search (sentence-transformers, 384-dim embeddings) and Reciprocal Rank Fusion
 - Implemented compressed inverted index with Variable-Byte encoding, Block-Max WAND early termination, and graph-filtered HNSW routing for pgvector
 - Deployed production stack on DigitalOcean with Docker, Nginx reverse proxy, Vercel frontend, and Cloudflare DNS/SSL
- ApnaTask — Hyperlocal Services Marketplace** | *PostgreSQL, PostGIS, WebSockets, Redis, Celery* Apr – Jun 2025
- Architected a dual-sided marketplace connecting service providers with customers using advanced PostGIS geospatial matching within configurable radius
 - Built real-time WebSocket-based negotiation engine and asynchronous notification system using Redis and Celery handling thousands of concurrent push notifications
- JobFlow AI — AI Resume Optimization & Job Discovery SaaS** | *React, TypeScript, Supabase, FastAPI* 2026
- Co-developed AI-powered resume builder with dual-flow (AI upload + manual), live premium templates, ATS fit scoring, and keyword intelligence for job matching
 - Built Python FastAPI scraper microservice that searches live web, classifies results with LLM, and extracts structured job data into Supabase
- WriteAI.me — AI Content Generation Platform** | *Python, FastAPI, Docker, Kubernetes, Stripe* 2023 – 2024
- Built high-performance AI writing SaaS with 80+ templates, multi-model orchestration, autonomous Agent Mode, and Monaco-based document editor serving copywriters on tiered subscriptions
- ZamDev AI CRM — Custom Lead & Project Management System** | *Next.js, TypeScript, Tailwind CSS* 2026
- Built a custom CRM from scratch for zamdevai.com to capture form submissions, manage lead pipeline (New, Contacted, Qualified, Proposal, Negotiating), track projects and tasks, and monitor analytics with real-time dashboard

Education

- Bachelor of Science (BS) in Artificial Intelligence** Oct 2024 – Jul 2028 (Expected)
- University of Central Punjab (UCP)* *Lahore, Pakistan*
- Lead Organizer, Google Developer Groups on Campus (GDGoC) UCP
 - Coursework: Data Structures, Database Systems, Algorithms, Artificial Intelligence, Machine Learning

Certifications

- Machine Learning Specialization** | *Stanford University & DeepLearning.AI (Andrew Ng)* Dec 2025
- Building AI Agents and Agentic Workflows Specialization** | *IBM* Dec 2025
- Agentic AI with LangGraph, CrewAI, AutoGen and BeeAI** | *IBM* Dec 2025
- Agentic AI with LangChain and LangGraph** | *IBM* Dec 2025
- Python for Data Science, AI & Development** | *Coursera* Mar 2025
- Self-Driving Car Specialization** | *Packt — Predictive Modeling, Embedded Systems* Dec 2025